

# CH 1 — Graphs of Scalar Functions

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## 1.0 Parametric Curves

We have seen functions that are either explicit or implicit

1.  $y = f(x)$
2.  $x = y^2$
3.  $x^2 + y^2 = 1$

We can also describe a curve using parametric equations where

$$x = x(t), y = y(t), t \in [a, b], \forall a, b \in \mathbb{Z}$$

with  $(x(t), y(t))$  is a point of the curve and  $t$  is the parameter.

- The curve is called **parametric curve**.
- We get a direction in which the curve is traced as well.

Practices:

1. Find the parametric curve:  $x = t - 1, y = 2t + 1, t \in [0, 3]$

$$x + 1 = t \implies y = 2(x + 1) + 1 = 2x + 3, x \in [-1, 2]$$

2. Find the parametric curve:  $x = \cos(t), y = \sin(t), t \in [0, 2\pi]$

$$x^2(t) + y^2(t) = \cos^2 t + \sin^2 t = 1, t \in [0, 2\pi]$$

3. Find the parametric curve:  $x = \cos(2t), y = \sin(2t), t \in [0, 2\pi]$

$$x^2 + y^2 = \cos^2(2t) + \sin^2(2t) = 1, t \in [0, 2\pi]$$

Same curve as Q2 but has two different parametrizations. Twice as the speed as the Q2.

## 1.1 Vector-Valued Functions

### Definition — Vector-Valued Function

It is a function whose domain is a subset of the  $\mathbb{R}$  and whose output is a vector

**Note:** a vector-valued function is a convenient way to collect scalar functions

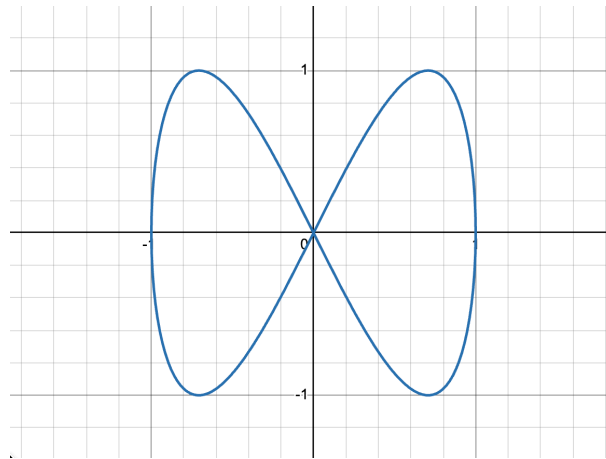
Examples:

1. In  $\mathbb{R}^2$ ,  $\vec{x}(t) = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$

2. In  $\mathbb{R}^3$ ,  $\vec{x}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix}$

3. Let  $\vec{x}(t) = (\cos t, \sin 2t)$  represent the position of a particle at time  $t$ . Sketch  $\vec{x}(t), 0 \leq t \leq 2\pi$

We have  $x = \cos t, y = \sin 2t, 0 \leq t \leq 2\pi$



### Definition – Velocity

$$\text{Velocity} = \vec{v}(t) = \frac{d\vec{x}(t)}{dt}$$

Examples:

1.  $\vec{x}(t) = (\cos t, \sin 2t)$

$$\text{velocity} = \frac{d\vec{x}(t)}{dt} = (-\sin t, 2 \cos 2t)$$

$$\text{At } t = 0, \vec{v}(0) = (0, 2) \quad \text{At } t = \frac{\pi}{2}, \vec{v}\left(\frac{\pi}{2}\right) = (-1, 0)$$

2. Let  $\vec{x}_1(t) = (\cos t, \sin t)$  and  $\vec{x}_2(t) = (\cos 3t, \sin 3t)$  be the paths of two particles, find the velocity of each particle when  $t = 0$

$$\frac{d\vec{x}_1(t)}{dt} = (-\sin t, \cos t), \text{ 1 revolution of unit circle radius 1 unit}$$

$$\frac{d\vec{x}_2(t)}{dt} = (-3 \sin 3t, 3 \cos 3t), \text{ 3 revolutions of unit circle radius 1 unit}$$

$$\vec{x}_1'(0) = (0, 1), \vec{x}_2'(0) = (0, 3)$$

### Definition – Tangent vector

Given the curve defined by

$$\vec{x}(t) = (x(t), y(t))$$

its **tangent vector** at  $t = t_0$  is

$$\vec{t} = \left. \frac{d\vec{x}}{dt} \right|_{t=t_0} = (x'(t_0), y'(t_0))$$

whenever  $\frac{d\vec{x}}{dt}, t = t_0 \neq \vec{0}$  exists.

**Note:**  $\vec{t}$  lies on the tangent line and points in the direction of travel

Example:

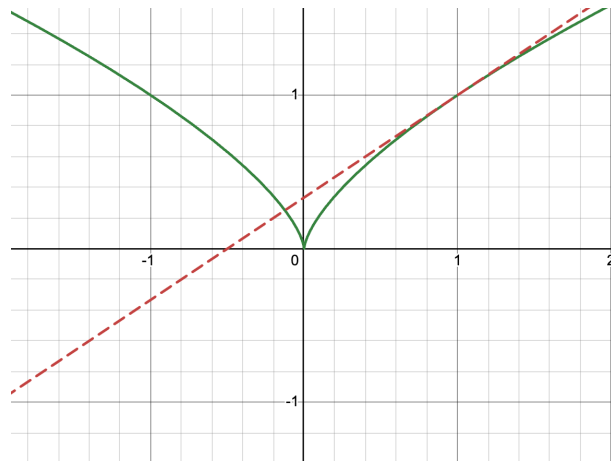
Consider the curve  $\vec{x}(t) = (t^3, t^2)$  and its tangent vector.

Notice that if  $x = t^3 \rightarrow t = x^{\frac{1}{3}} \rightarrow y = x^{\frac{2}{3}}$

$$\frac{d\vec{x}(t)}{dt} = (3t^2, 2t)$$

$$t = 1, \vec{x}(1) = (1, 1), \vec{v}(1) = (3, 2)$$

$t = 0, \vec{x}(0) = (0, 0), \vec{v}(0) = (0, 0) = \vec{0}$  We cannot define the tangent vector as the function at the point is not differentiable



## 1.2 Scalar Functions

Recall:

$f : A \rightarrow B$  is a function

1. Each element  $a \in A$  is a unique element  $f(a) \in B$  called the **image** of  $a$  under  $f$
2. The set  $A$  is called the **domain** of  $f$  and is denoted by  $D(f)$
3. The set  $B$  is called the **codomain** of  $f$ .
4. The subset of  $B$  consisting of all  $f(a)$  is called the **range** of  $f$ , denoted by  $R(f)$
5. Single variable scalar functions are  $f : \mathbb{R} \rightarrow \mathbb{R}$

### Definition – Scalar Function

A **scalar function**  $f(x_1, \dots, x_n)$  of  $n$ -variables is a function whose domain is a subset of  $\mathbb{R}^n$  and whose range is a subset of  $\mathbb{R}$

Examples:

1.  $f(x, y) = x^2 + y^2 \rightarrow f(1, 1) = 1^2 + 1^2 = 2$
2.  $f(x, y) = \sin(x + y) \rightarrow f(0, \frac{\pi}{2}) = \sin(\frac{\pi}{2}) = 1$
3.  $f(x, y, z) = e^{x+y+z} \rightarrow f(1, 2, 1) = e^{1+2+1} = e^4$

**Notation:** We can write  $\vec{x} = (x_1, x_2, \dots, x_n)$  so that  $f(\vec{x}) = f(x_1, x_2, \dots, x_n)$

Our independent variables don't have to be  $x, y, z$ , etc... Multivariable functions can be used to model many different physical situations

Example: write down a multivariable function which calculates the volume of a cylinder.

$$V = \pi r^2 h \rightarrow V(r, h) = \pi r^2 h$$

### Definition – Domain and Range

**Domain** is the largest subset of  $\mathbb{R}^n$  for which the function is defined

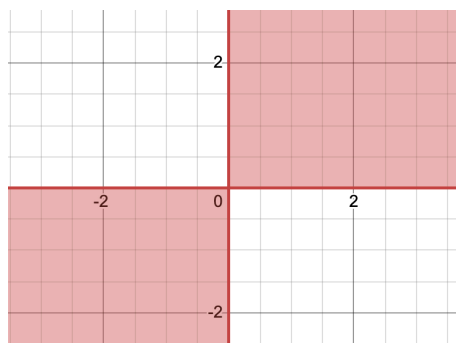
**Range** is a set of all possible values which are subset of  $\mathbb{R}$

Examples:

1. Find the domain and range of  $f(x, y) = \sqrt{xy}$

$$\text{We need } xy \geq 0 \Rightarrow \begin{cases} x \geq 0 \text{ and } y \geq 0 \\ x \leq 0 \text{ and } y \leq 0 \end{cases}$$

$$\text{dom}(f) = \{(x, y) \in \mathbb{R}^2 : x \geq 0, y \geq 0\} \cup \{(x, y) \in \mathbb{R}^2 : x \leq 0, y \leq 0\}$$



We know that  $\sqrt{xy} \geq 0, \forall x, y \in \text{dom}(f)$

Let  $c \in \mathbb{R}, c \geq 0$ , be an arbitrary non-negative real number.

Notice that  $f(c^2, 1) = \sqrt{c^2} = |c| = c$  since  $c \geq 0$

For any  $c \in \mathbb{R}, c \geq 0, \exists$  inputs in the domain of  $f$  s.t.  $c$  is a valid output  $\rightarrow c \in \text{ran}(f) : \mathbb{R}_{\geq 0}$

2. Find the domain and range of  $g(x, y) = \frac{x^2 - y^2}{|x| + |y|}$

We need  $|x| + |y| \neq 0$  which is true  $\forall x, y \in \mathbb{R}$  both not zero since  $|x| > 0, |y| > 0$

$$\text{dom}(g) = \{(x, y) : x, y \in \mathbb{R} \setminus \{(0, 0)\}\}$$

Consider case  $x > y: x^2 - y^2 > 0, |x| + |y| > 0 \Rightarrow g(x, y) > 0$

Consider case  $x = y: x^2 - y^2 = 0, |x| + |y| = 2x \Rightarrow g(x, y) = 0$

Consider case  $x < y: x^2 - y^2 < 0, |x| + |y| > 0 \Rightarrow g(x, y) < 0$

or alternatively

$$\text{Let } c \in \mathbb{R}, c > 0, g(c, 0) = \frac{c^2 - 0}{|c| + 0} = |c| = c, c > 0$$

$$\text{Let } d \in \mathbb{R}, d < 0, g(0, d) = \frac{0 - d^2}{0 + |d|} = -|d| = d, d < 0$$

$$\text{Let } a, b \in \mathbb{R}, a = b, g(a, b) = \frac{a^2 - b^2}{|a| + |b|} = \frac{a^2 - a^2}{|a| + |a|} = \frac{0}{2a} = 0$$

Thus  $\text{ran}(g) = \mathbb{R}$

### 1.3 Geometric Interpretation of $z = f(x, y)$

Question: How we can graph  $z = f(x, y)$

ANS: We graph  $(a, b, f(a, b))$ . We think of  $f(a, b)$  as height above the  $xy$ -plane

#### Definition – Level Curves

The **level curves** of a function  $f(x, y)$  are the curves

$$f(x, y) = k$$

where  $k$  is a constant in the range of  $f$

- **Alternate definition:** curves along which the height of the graph  $z = f(x, y)$  is constant
- A collection of **level curves** are called a **contour map**
- There are **exceptional level curves** because it looks different from the other level curves.

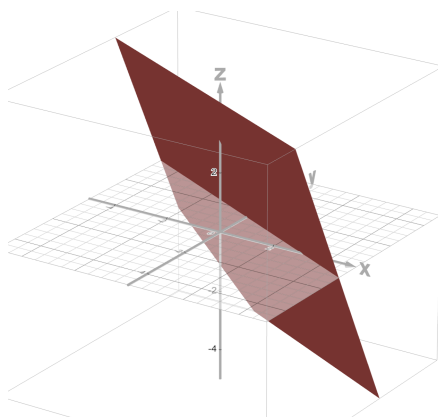
#### Definition – Cross Sections

A **cross section** of a surface  $z = f(x, y)$  is the intersection of  $z = f(x, y)$  with a plane

Examples:

1. Sketch  $x + 2y + z = 3$

We can rewrite this as  $z = 3 - x - 2y = f(x, y)$

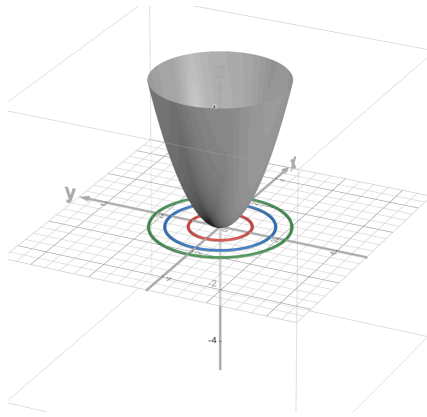


Sketch and write down an equation which is the level curves for the above and sketch these level curves for a different  $k$

Level curve:  $3 - x - 2y = k, k \in \mathbb{R} \implies y = -\frac{1}{2}x + \frac{1}{2}(3 - k)$

2. Sketch  $f(x, y) = x^2 + y^2$

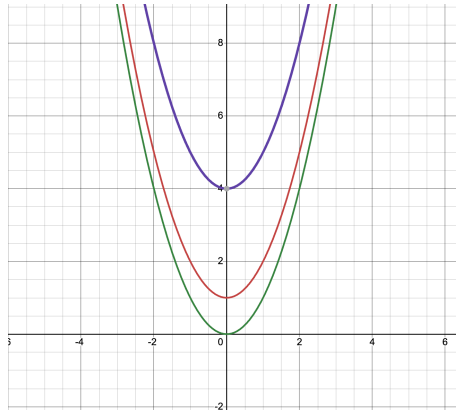
Level curves:  $x^2 + y^2 = k, k \geq 0$  (red circle below)



This is a paraboloid (grey “cone-like” shape)

3.  $z = x^2 + y^2$

$$x = c \implies z = c^2 + y^2$$

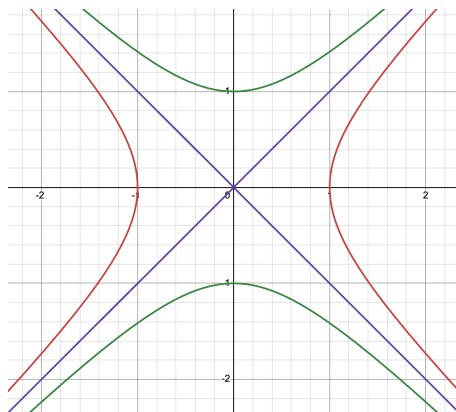


$$y = d \implies z = x^2 + d^2 \text{ (Similar like above)}$$

4.  $z = x^2 - y^2$

$$\text{dom}(f) = \mathbb{R}, \text{ran}(f) = \mathbb{R}$$

Level curves:  $x^2 - y^2 = k, k \in \mathbb{R}$  (hyperbolas)



Red lines  $k > 0$

Green lines  $k < 0$

Blue lines  $k = 0$

 **Definition – Levels**

1. A **level surface** of a scalar function  $f(x, y, z)$  is defined by

$$f(x, y, z) = k, \quad k \in R(f)$$

2. A **level set** of a scalar function  $f(x), x \in \mathbb{R}^n$  is defined by

$$f(x) = k, \quad k \in R(f)$$